Astronomy - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What drives fusion?
   (a) Spontaneous emission from radioactive atoms.
   (b) The minimization of gravitational potential energy.
   (c) Thermally induced collisions.
   (d) Plate tectonics.
2. Why do we see only the same face of the Moon at all times?
   (a) because the Moon has a nearly circular orbit around the Earth
   (b) because the Moon does not rotate
   (c) because the other face points toward us only at new moon when we can't see the Moon
   (d) because the Moon's rotational and orbital periods are equal
3. Previous IAAC rounds featured Proxima/Alpha Centauri as closes star(system) to the Earth. Which one is the second closest star(system)?
   (a) Wolf 359
   (b) Sirius
   (c) 61 Cygni
   (d) Barnard's Star
4. Why is the sky red?
   (a) Because the molecules that compose the Earth's atmosphere have a blue-ish color.
   (b) Because the sky reflects the color of the Earth's oceans.
   (c) Because the atmosphere preferentially scatters short wavelengths.
   (d) Because the Earth's atmosphere preferentially absorbs all other colors.
5. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.
   (a) It has a fusion crust
   (b) It contains solidified spherical droplets
   (c) It is highly processed
   (d) It has different elemental composition than earth

True / False (10 marks)

Answer True or False

6. True or False: The Sun mainly emits white light, which causes the sky to appear blue during the day.
7. True or False: The surface of Venus is approximately 4.5 billion years old, suggesting a much older geological formation than is accurate.
8. True or False: Venus lacks seasons because its rotation axis is nearly parallel to the plane of the Solar System.
9. True or False: Bolometric luminosity in astronomy is the luminosity integrated only over all wavelengths of light.
10. True or False: Saturn's equatorial wind speed is greater than 900 miles per hour, unlike Jupiter's high-speed winds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of observing a hydrogen spectral line at 486.3 nm in a star's spectrum, compared to the known laboratory wavelength of 486.1 nm. What does this indicate about the star's motion?
12. Discuss the categories of electromagnetic radiation, analyzing their wavelengths from shortest to longest, and explain the implications of this order for astronomical observations.

End of Paper